

REVIEW



Macrophage memory: Types, mechanisms, and its role in health and disease

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Abstract

On the basis of the mechanisms of action and characteristics of immune effects, immunity is commonly categorized into innate and adaptive immunity. Adaptive immunity is associated with the response to non-self-entities and is characterized by high specificity and memory properties. In contrast, innate immunity has traditionally been considered devoid of memory characteristics. However, an increasing number of studies have sought to challenge this conventional immunological dogma and shown that innate immune cells exhibit a more robust and rapid response to secondary stimulation, thus providing evidence of the immunological memory in innate immunity. Macrophages, which are among the most important innate immune cells, can also acquire memory phenotype that facilitates the mediation of recall responses. Macrophage memory is a relatively new concept that is revolutionizing our understanding of macrophage biology and immunological memory and could lead to a new class of vaccines and immunotherapies. In this review, we describe the characteristics and mechanisms of macrophage memory, as well as its essential roles in various diseases.

KEYWORDS

endowed immunity, epigenetic reprogramming, macrophage memory, trained immunity

INTRODUCTION

Depending on the types of participating cells, recognition patterns, and effector mechanisms, immunity is traditionally divided into two categories, namely, innate immunity and

adaptive immunity. The former is mediated by phagocytes, such as monocytes, macrophages, and neutrophils, as well as natural killer cells (NK cells), and is characterized by a rapid and non-specific immune response, whereas the latter is mainly mediated by T and B lymphocytes and is characterized by a slower, albeit highly specific response. It is generally believed that the memory properties are associated exclusively with adaptive immunity and not innate immunity. Nevertheless, it has long been debated as to whether innate immunity also has memory characteristics, and the recent findings have indeed indicated that immunological memory is not an exclusive hallmark of adaptive immunity [1, 2]. For example, plants and invertebrates, which lack adaptive immunity, are protected against secondary infections, and invertebrates can also exhibit transplant rejection [3–6]. Macrophages, which are key innate immune cell types, participate in the innate immunity by phagocytizing

Abbreviations: AMs, alveolar macrophages; BCG, Bacillus Calmette-Guérin; DAMPs, danger-associated molecular patterns; H3K4me3, tri-methylation of histone H3 at lysine 4; HIF-1 α , hypoxia-inducible factor-1 α ; HSCs, haematopoietic stem cells; IL, interleukin; lncRNAs, long non-coding RNAs; LPS, lipopolysaccharide; MCP-1, monocyte chemoattractant protein 1; MHC, major histocompatibility complex; miRNAs, microRNAs; mTOR, mammalian target of rapamycin; NK, natural killer; ox-LDL, oxidized low-density lipoprotein; PAMPs, pathogen-associated molecular patterns; PBMCs, peripheral blood mononuclear cells; PIR-A, paired immunoglobulin-like receptor-A; PRRs, pattern recognition receptors; SCID, severe combined immune-deficiency; TAM, tumour-associated macrophage; TB, tuberculosis; TLR, toll like receptor; TNF- α , tumour necrosis factor- α .

TABLE 1 Differences in characteristics of innate immune memory and adaptive immune memory.

Characteristics	Innate immune memory		Adaptive immune memory
	Trained immunity	Endowed immunity	
Memory	Yes		Yes
Organism	Plants, invertebrate and vertebrates		Higher vertebrates
Cells	Monocytes/macrophages, NK cells, granulocytes, innate lymphoid cells		T and B lymphocytes
Lasting time	Weeks to months		Years
Stimulators	Innate ligands (LPS, BCG and β -glucan)	Allogeneic antigens and respiratory viral infection	Antigens
Receptors	PRRs	PIR	TCR, BCR
Help of T cells	Independent	Essential	–
Molecular mechanisms	Epigenetic reprogramming and cell metabolic change	Unknown	Antigen specific gene rearrangement
Specificity	Non-specific/semi-specific	Yes	Yes

Abbreviations: BCG, Bacillus Calmette-Guérin; BCR, B cell receptor; LPS, Lipopolysaccharide; NK, Natural killer; PIR, Paired immunoglobulin-like receptor; PRRs, Pattern recognition receptors; TCR, T cell receptor.

foreign bodies and initiate the adaptive immunity via capturing and processing antigens, which are subsequently presented to lymphocytes, thereby playing a central role in immune responses [7, 8]. Moreover, macrophages exhibit certain immunological memory-related properties that are traditionally attributed to adaptive T and B cells. However, compared with the memory properties of adaptive immunity, macrophage memory remains comparatively unknown. Herein, we describe two types of macrophage memory, namely, trained immunity (innate immunological memory) and endowed immunity (adaptive immune-like memory), as well as the essential roles these play in a range of diseases. Moreover, the differences in characteristics of innate immune memory and adaptive immune memory are shown in Table 1.

TRAINED IMMUNITY (INNATE IMMUNOLOGICAL MEMORY)

In 2011, Netea et al. [3] first proposed the concept of ‘trained immunity’ to describe the ability of mammalian innate immunity to exhibit immunological memory, monocytes/macrophages, NK cells and granulocytes show altered immune responses to secondary stimulation originating from exogenous or endogenous insults. In terms of cell populations and molecular mechanisms, there are important differences between trained immunity and classic adaptive immune memory. Adaptive immune memory generally implies high specificity and cascade amplification, while trained immunity is a non-specific or semi-specific phenomenon that is mainly characterized by alterations in the number and/or function of innate immune cells, and subsequently involves the initiation of a pro-inflammatory

or anti-inflammatory secondary immune response [3, 9, 10]. Depending on the nature and the dose of the primary stimulation, trained immunity responds to the secondary stimulation through two opposite functional programmes, with enhanced immunological imprinting being manifested as training, and the suppressed programme contributing to tolerance (Figure 1) [10, 11].

Discovery and characteristics of trained immunity

Freudenberg et al. [12] reported that in mice, macrophages primarily stimulated by sepsis or high-dose lipopolysaccharide (LPS) induced immune tolerance and exhibited a suppressed immune response to avoid the development of a state of excessive inflammation on secondary challenge. Quintin et al. [13] also showed that when macrophages were trained by β -glucan (the main cell wall structural component of the yeast *Candida albicans*), mice lacking functional T and B lymphocytes were protected against reinfection of multiple pathogens such as *C. albicans* and *Staphylococcus aureus*. Indeed, an increasing number of studies have indicated that bacterial, fungal, parasitic, and viral infections can train monocytes/macrophages to promote the secretion of inflammatory cytokines and enhance phagocytosis in response to secondary stimulation, thereby confirming the universality of macrophage memory [13–17]. In addition to studies using animal models, a study on childhood vaccination reported that the Bacillus Calmette-Guérin (BCG) vaccination enhanced the defence against infections in a monocyte-dependent manner, thereby leading to an overall reduction in child mortality [18]. Furthermore, accumulating

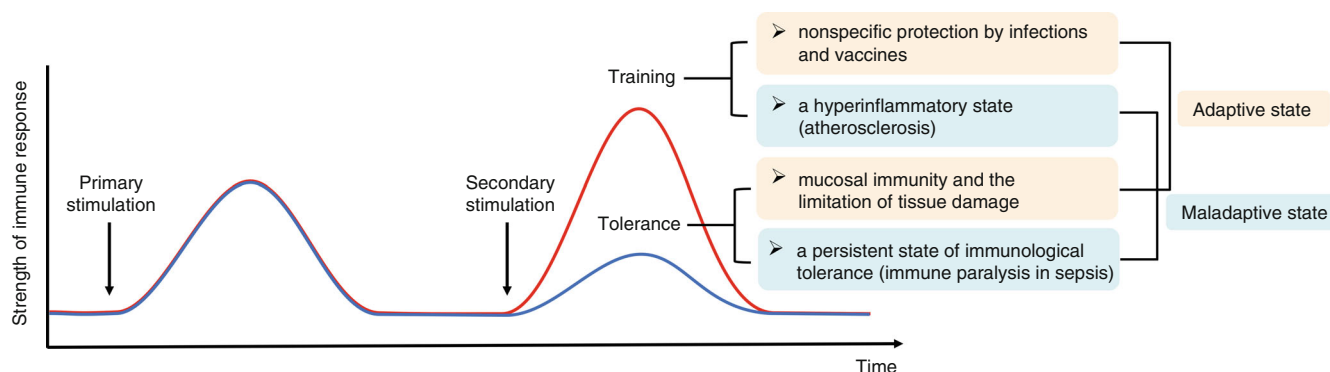


FIGURE 1 Two opposite functional programmes of trained immunity: training programme and tolerance programme. Innate immune cells acquire a memory phenotype and respond to secondary stimulation with qualitatively and quantitatively adjustment, where the enhanced secondary immune response is training programme and the suppressed immunological imprinting is tolerance programme. Training and tolerance programmes are both two-sided. On the one hand, training programme favours nonspecific protection from infections and vaccines and tolerance programme is beneficial for mucosal immunity and the limitation of tissue damage. On the other hand, these programmes may disrupt homeostasis and contribute to disease progression, leading to either a hyperinflammatory state or a persistent state of immune tolerance.

epidemiological data indicated that live vaccines, such as the measles, smallpox, and polio vaccines, provide non-specific protective effects against a broad range of infections other than those caused by the respective target viruses [19–22].

These studies have revealed two important characteristics of trained immunity. Firstly, trained immunity is a non-specific or semi-specific phenomenon, with its specificity being lower than that of adaptive immunity; and as such, it can provide cross-protection against a wide range of pathogens. Secondly, trained immunity primarily stimulated by infections or vaccines induces a pro-inflammatory or anti-inflammatory secondary immune response independent of classical T/B cell adaptive immunity.

Mechanisms of trained immunity

Unlike the classical adaptive immune memory, which is dependent on antigen-specific gene rearrangement, the mechanisms underlying trained immunity mainly include (1) altered pattern recognition receptors (PRRs) expression, (2) epigenetic reprogramming, and (3) metabolic reprogramming (Figure 2). Moreover, these mechanisms may be characterized by a degree of synergism, rather than functioning completely independently.

Altered PRRs expression

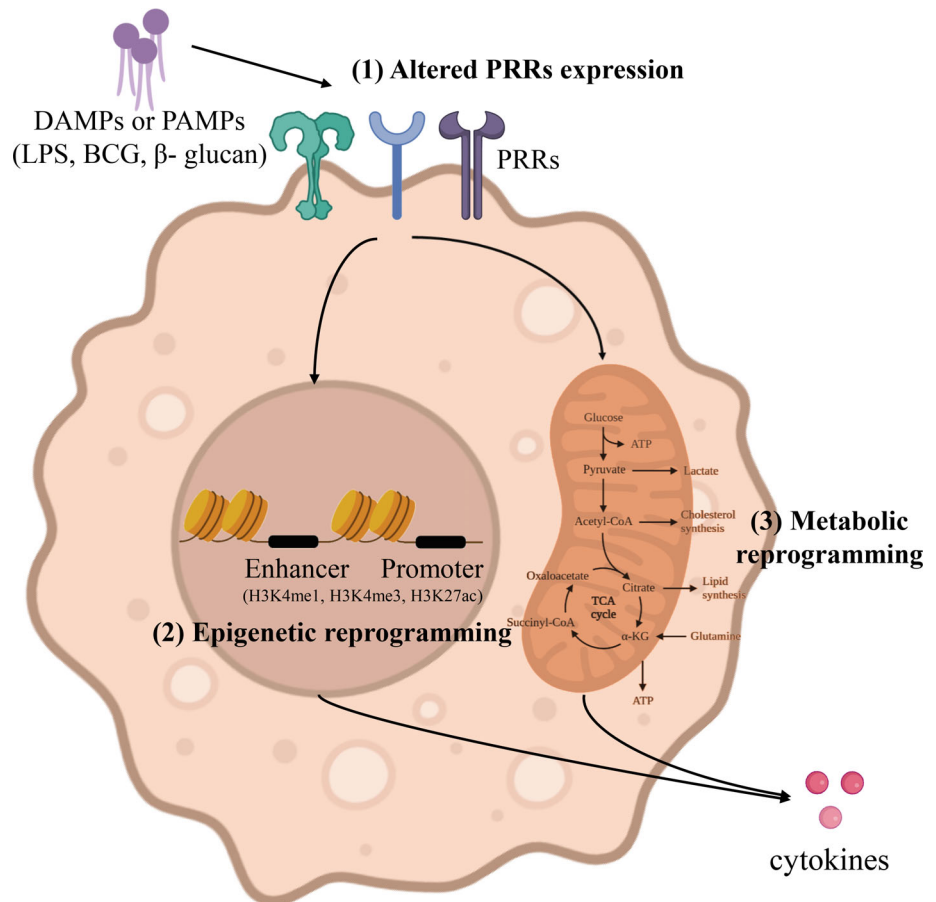
In response to the primary stimulation, macrophages acquire a memory phenotype and upregulate PRRs expression to promote the recognition of diverse foreign

antigen-derived pathogen-associated molecular patterns (PAMPs) and self-derived danger-associated molecular patterns (DAMPs), and subsequently induce an altered secondary response [23–25]. Following stimulation by different PRRs ligands such as the dectin-1 ligand β -glucan, Toll-like receptor 2 (TLR2) ligand lipoteichoic acid, TLR4 ligand LPS, and nucleotide-binding oligomerization domain-containing protein 2 (NOD2) ligand muramyl dipeptide, monocytes/macrophages modify the secretion of pro-inflammatory cytokines such as tumour necrosis factor- α (TNF- α) and interleukin (IL)-6, and initiate an altered immune response on secondary encounter [26]. In addition, Macrophage memory induced by cell surface receptors may be retained following cell differentiation. The findings of a recent study have revealed that macrophages derived from mice or human haematopoietic stem and progenitor cell subsets that were exposed to a TLR2 agonist prior to or during macrophages differentiation developed immune tolerance during the secondary response and produced lower levels of the inflammatory cytokines TNF- α , IL-6 and IL-1 β [27].

Epigenetic reprogramming

Epigenetic reprogramming plays a crucial role in the regulation of gene expression through the alteration of chemical modifications in DNA and histone molecules within chromatin. This process is characterized by its dynamic and reversible nature, enabling the generation of heritable phenotypic changes without altering the DNA sequence [28]. As the primary mechanism underlying trained immunity, epigenetic reprogramming encompasses various processes

FIGURE 2 Mechanisms of trained immunity. (1) Altered PRRs expression. The expression of receptors on the cell surface is altered to improve pathogen recognition. (2) Epigenetic reprogramming. The fundamental cause of trained immunity. It can produce heritable phenotypic changes without altering the DNA sequence, mainly including histone modifications, DNA methylation, miRNAs and lncRNAs. (3) Metabolic reprogramming. Innate immune cells change their metabolic patterns to accommodate host responses. For example, glucose metabolism is shifted from oxidative phosphorylation to glycolysis.



such as histone modification, DNA methylation, microRNAs (miRNAs) and long non-coding RNAs (lncRNAs).

In the case of *Candida* infections, monocytes/macrophage memory trained by β -glucan has been found to be associated with epigenetic changes involving the enhanced tri-methylation of histone H3 at lysine 4 (H3K4me3) [13]. The H3K4me3 histone marker is located on the promoter regions of IL-6, TNF- α and IL-1 β . Its increased expression promoted the release of pro-inflammatory cytokines, thereby enhancing the response to secondary stimulation and aiding in the fight against infection [13, 29, 30]. Moreover, in vitro stimulation of human monocytes trained by oxidized low-density lipoprotein (ox-LDL) using TLR-2 and TLR-4 agonists leads to upregulation of atherosclerosis-related cytokines such as IL-6, IL-8, IL-18 and TNF- α , as well as monocyte chemoattractant protein 1 (MCP-1), matrix metalloproteinase (MMP) and the promoter marker H3K4me3 [31]. Kleinnijenhuis et al. [30] also reported that the protective effect of monocytes against secondary reinfection after primary BCG vaccination in healthy volunteers was mediated by H3K4me3, accompanied by the production of the inflammatory cytokines IFN- γ , TNF and IL-1 β increasing several-fold in response to infections by non-specific bacterial and fungal pathogens. Moreover, the inhibition of histone methyltransferase was found to be

associated with a significant reduction in BCG-induced macrophage memory.

On the basis of the initial inflammatory status (the expression of pro-inflammatory mediators, such as TNF- α and IL-6), LPS exposure and β -glucan priming have been shown to induce two opposite functional programmes of macrophage memory: training programme and tolerance programme. The opposite functional changes in macrophages are attributed to specific epigenetic changes such as H3K4me1 and H3K27ac. H3K27ac, which marks active promoters and enhancers, is in a dynamic modification that gradually disappears after the stimulation subsided, and the distal regulatory element (enhancer) marker H3K4me1 remains accessible [11]. Additionally, when macrophages are stimulated with low-dose LPS, latent enhancers which are normally inactive, label-free, and unbound by transcription factors are selectively activated and acquired H3K4me1 marks after the resolution of stimulation, and once having been unveiled, the histone marks persist and mediate a stronger pro-inflammatory response upon secondary stimulation [32]. However, when monocytes/macrophages are stimulated in response to high-dose LPS, the induction of macrophage memory involves repressive histone H3K9me2 and H3K27me2 modification, leading to reduced expression of pro-inflammatory cytokines and a suppressed secondary response [11, 33, 34].

Recent studies have shown that DNA methylation also contributes to the epigenetic reprogramming of macrophage memory. For instance, a stable and robust differential DNA methylation pattern was observed in peripheral blood mononuclear cells (PBMCs) isolated from BCG-vaccinated individuals compared to counterparts without BCG vaccination. Additionally, gene ontology analysis revealed that promoters with altered DNA methylation patterns were strongly enriched among immune-related genes, thereby enhancing the resistance of macrophages to mycobacteria [35, 36]. Indeed, a global methylation analysis showed that there existed over 1000 differentially methylated regions between PBMCs of tuberculosis (TB) patients and healthy controls, and there were still almost 4000 differentially methylated regions following completion of anti-TB treatment, and the majority of the affected genes were associated with the autophagy pathway [37, 38]. Similarly, a distinct DNA methylation profile, which was enriched in the pentose phosphate pathway, T cell migration and IFN- γ production pathways, was uncovered in alveolar macrophages (AMs) after *M. tuberculosis* infection [39]. Intriguingly, there were more differentially methylated regions in AMs than in PMBCs within the same individual [38]. In addition, altered DNA methylation pattern associated with the immune gene loci can also be detected in the sperm DNA of parental male mice infected with fungus *C. albicans* [40].

Further factors that play important roles in the induction of macrophage memory are miRNAs and lncRNAs. MiRNAs, which serve as key regulators of immune cell development and function, can persist after the primary stimulation, and trigger a stronger secondary response due to their long half-life [41, 42]. In this regard, Seeley et al. [43] have reported that prolonged exposure to LPS in mice led to an increase in the expression of miRNA-221 and miRNA-222 in bone marrow-derived macrophages, followed by transcriptional silencing of certain inflammatory genes through switch/sucrose non-fermentable (SWI/SNF) and signal transducer and activator of transcription (STAT)-mediated chromatin remodelling. This in turn was found to promote immune tolerance to secondary challenge, thereby indicating the regulatory effects of miRNAs in the functional reprogramming of macrophage memory. An enhancement in the expression of miRNA-221 and miRNA-222 has also been established to be associated with immunoparalysis and heightened organ damage in patients with sepsis. Moreover, the upregulation of miRNA-155 has also been reported to be correlated with excessive activation of myeloid cells in inflammatory responses, and this may be attributed to a decreased activity of phosphatase SHIP1, which has recently been shown to act as a negative regulator in inducing trained immunity [44, 45]. In addition to miRNAs, it is conceivable that lncRNAs also play central roles in macrophage memory. Indeed, Fanucchi et al. [46] found that the newly

identified immune gene initiation lncRNA (IPL) directed WD repeat-containing protein 5 (WDR5)-mixed lineage leukaemia protein 1 (MLL1) histone methyltransferase complex formation and subsequent H3K4me3 accumulation on the promoters of IL-6, IL-8, and CXCL1, thereby resulting in an enhanced inflammatory response in β -glucan-induced trained immunity.

Moreover, systematic analysis of macrophage subsets indicated that environmental signals also appeared to play significant roles in determining macrophage phenotype by regulating the expression and function of transcription factors that activated cis-acting enhancer elements [47–49]. Indeed, environmental signals regulated the macrophage memory through altered histone modification in promoters and enhancers, resulting in long-lasting effects on the transcriptional profile and inflammatory potential of monocytes/macrophages [50].

Metabolic reprogramming

Trained immunity has also been established to involve changes in cell metabolism, as indicated by the different metabolic pathways of two types of activated macrophages, namely the classically activated M1 macrophages and alternatively activated M2 macrophages. Whereas M1 macrophages tend to utilize glycolytic metabolism for energy generation, M2 macrophages are primarily dependent on oxidative phosphorylation for ATP biogenesis [51, 52]. In this context, Cheng et al. [53] have reported that the metabolic basis of trained immunity triggered by the *C. albicans* cell wall constituent β -glucan was the induction of aerobic glycolysis dependent on an AKT-mammalian target of rapamycin (mTOR)-hypoxia-inducible factor-1 α (HIF-1 α) pathway. Furthermore, trained human monocytes have been shown to be characterized by high glucose consumption, high lactate production, and a high ratio of nicotinamide adenine dinucleotide (NAD⁺) to nicotinamide adenine dinucleotide (NADH), reflecting a shift in cell metabolism from oxidative phosphorylation to aerobic glycolysis. Similar changes in glucose metabolism have also been observed in BCG-induced trained immunity [54]. Such changes in cell metabolism facilitated the rapid conversion of glucose to lactic acid in the cytoplasm, thereby yielding the energy necessary to mount a rapid response to secondary challenge [55]. Furthermore, in the case of LPS-induced immune memory, macrophages have been observed to display strong and transient glycolytic metabolism during the acute response, whereas after the stimulation subsided, the metabolic polarity shifted back to oxidative phosphorylation, and histones in the transcriptional region of the genome were deacetylated by deacetylase-1 and deacetylase-6, thereby establishing the immune tolerance memory of macrophages [56]. In

addition, intermediate metabolites have also been established to be involved in the regulation of macrophage memory. For example, the metabolite mevalonate has been shown to induce trained immunity by activating IGF1-R and mTOR and subsequent histone modifications in inflammatory pathways [57]. Furthermore, the tricarboxylic acid cycle (TCA) metabolite fumarate provided a link integrating immune and metabolic circuits to induce the epigenetic reprogramming of monocytes by downregulating KDM5 histone demethylases [58, 59]. And glutamine metabolism has been demonstrated to promote a high accumulation of α -ketoglutarate, and a corresponding elevation in the α -ketoglutarate/succinate ratio was found to be sufficient to regulate H3K27me3 modification and ten-eleven translocation-dependent DNA demethylation, which contribute to the induction of macrophage memory [58–60].

Endowed immunity (adaptive immune-like memory)

However, recent studies also provide a fundamental basis for challenging the assumption that innate immune cells acquire a memory phenotype independent of adaptive

immunity. In 2022, Wu et al. proposed the concept of ‘endowed immunity’ (also known as adaptive immune-like memory) to elucidate the phenomenon that macrophage memory requires the help of adaptive immunity [61]. Similarly, in endowed immunity, macrophages are endowed with adaptive immune-like memory that facilitates the recall of primary immune responses, however, high-specificity endowed immunity responses to secondary stimulation necessitate the assistance of helper T cells (Figure 3) [61].

Discovery and characteristics of endowed immunity

Dai et al. [62] showed that mice monocytes/macrophages acquired an alloantigen-specific memory in response to secondary encounters, and identified that paired immunoglobulin-like receptor-A (PIR-A), a major histocompatibility complex (MHC)-I receptor, was necessary for the induction of macrophage memory. Either deleting PIR-A or blocking the PIR-A-MHC-I combination was demonstrated to suppress memory and attenuate the allograft rejection in kidney and heart transplantation.

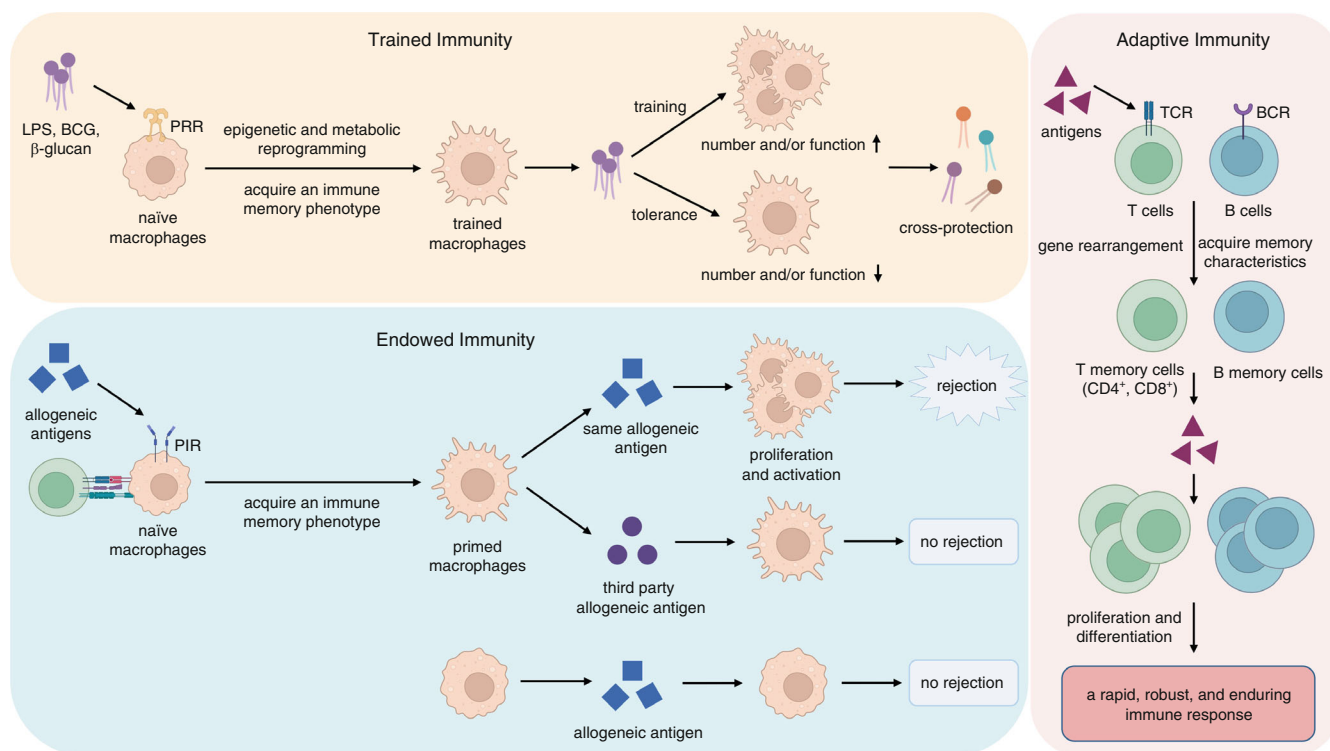


FIGURE 3 Two types of macrophage memory: trained immunity and endowed immunity. In trained immunity, macrophages acquire an immune memory phenotype via epigenetic and metabolic reprogramming and exhibit an enhanced (training) or suppressed (tolerance) host defence response towards the secondary stimulation. In endowed immunity, macrophages primed by allogeneic antigens acquire the potential ability to reject allogeneic grafts bearing the same antigen, with the assistance of helper CD4⁺ T cells, although lack the capacity to reject third-party grafts. In adaptive immunity, T and B cells display a highly specific immune (memory subsequent to primary exposure to antigens and produce a rapid, robust, and enduring immune response.

Moreover, Liu et al. [63] have reported that macrophages had the capacity to recognize and reject allogeneic antigens during secondary stimulation, which requires two signals, primed by allogeneic antigens and the assistance of CD4⁺T cells.

Similarly, Chu et al. [64] also showed that macrophages primed with allogeneic antigens mounted an acute response to skin allografts with a certain degree of antigen specificity, which was similar to the characteristics of adaptive immune cells. First, the authors demonstrated that immunodeficient recipient mice showed no signs of rejection to allogeneic skin grafts, which was consistent with the findings of previous studies that have established that innate immune cells alone were insufficient to facilitate the rejection of allogeneic organ grafts. In addition, the authors further found that immunodeficient recipient mice, which received antigen-immunized macrophages (primed macrophages) from immunocompetent mice, were able to efficiently reject the same allogeneic skin grafts, although not third-party skin grafts, or showed significantly slower rejections of the grafts. In summary, when primed macrophages in immunocompetent mice were exposed to the same allogeneic antigen secondarily, primed macrophages underwent pronounced proliferation, expanded during the immune response phase and exerted a direct graft-rejection effect, at least partially, via a perforin-dependent pathway. Moreover, the assistance of CD4⁺T cells, but not CD8⁺T cells, during the priming period was found to be essential for naïve macrophages to gain the ability to distinguish allografts and thus promote rejection [64–66]. In a follow-up study, the same authors reported that the specific memory of primed macrophages in the rejection of allogeneic cells and skin grafts was long-term and persisted for at least 4 months (until the end of the experimental period) [67]. Accordingly, suppressing this memory may represent a potential therapeutic strategy for improving transplantation outcomes.

Furthermore, Yao et al. [68] found that the induction of memory AMs required the assistance of effector CD8⁺T cells, but not CD4⁺T cells. Firstly, the authors demonstrated that respiratory viral infection induced lasting alveolar macrophage memory. Subsequently, CD4⁺T and/or CD8⁺T cells were depleted *in vivo* by repeated injections of anti-CD4 and/or CD8 monoclonal antibodies, and the authors found that CD8⁺T cells were necessary for the induction of memory AMs while the depletion of CD4⁺T cells alone failed to do so. Afterwards, severe combined immune-deficiency (SCID) mice, which lack functional T cells and B cells, were designed to further verify the conclusion. Similarly, infected SCID mice failed to induce alveolar macrophage memory, but transferring CD8⁺T cells to the airways of infected SCID hosts restored the immunological memory phenotypically and functionally.

More importantly, the authors further demonstrated that the assistance of CD8⁺T cells was required for the priming, but not maintenance, of memory AMs.

Taken together, the findings of the aforementioned studies indicate that functional endowed immunity requires the assistance of different types of T cells in different disease models. Hence, further studies are needed to determine which T cell subsets facilitate the induction of macrophage memory under different conditions. The precise nature of the underlying molecular mechanisms for endowed immunity is not yet unclear. Some studies decipher the phenomenon through the mechanisms of trained immunity [64, 68], while others suggest that endowed immunity may be associated with the secretion of inflammatory effectors and amplified adaptive immune responses [61, 69].

Central versus peripheral macrophage memory

Macrophage memory was initially shown to act through mature myeloid cells. However, this hypothesis also poses a problem due to the short lifespan of monocytes/macrophages in circulation [70, 71]. Therefore, how macrophage memory is maintained for weeks, months or even years remains unknown [22]. Recent studies have proposed a new perspective and suggested that immunological memory can occur not only in blood monocytes and tissue macrophages (peripheral macrophage memory) but also in bone marrow progenitor cells (central macrophage memory).

In a mouse model of tuberculosis, Kaufmann et al. [72] found that BCG vaccination reprogrammed the transcriptional landscape of haematopoietic stem cells (HSCs) and multipotent progenitors (MPPs) in an IFN- γ -dependent manner, resulting in enhanced myelopoiesis and local cell proliferation. What is more, macrophages generated by BCG-educated HSCs provided a more effective defence against deadly mycobacterium tuberculosis infections. Similarly, β -glucan priming was reported to induce trained immunity via modulation of CD41⁺ HSCs and MPP3, which were closely related to the elevated IL-1 β and granulocyte-macrophage colony-stimulating factor (GM-CSF) signalling, as well as the adaptations in glucose metabolism and cholesterol biosynthesis [73]. In addition, it has been established that embryonic-derived tissue-resident macrophages were the source of memory AMs in endowed immunity after respiratory viral infection. The authors found that memory AMs developed and self-sustain independently of circulating monocytes and bone marrow progenitor cells through assessing the expression of Siglec-F (embryonic) and CCR2 (monocytic) and the use of CCR2^{-/-} mice, as well as the BM chimeric approach [68].

The essential role of macrophage memory in disease responses

The memory macrophages response to the secondary stimulation with enhanced or suppressed immunological imprinting and is closely associated with various diseases. Macrophage memory is a double-edged sword. On the one hand, as a novel strategy for disease prevention and treatment, macrophage memory is beneficial for promoting host immune response against infections. On the other hand, this type of memory may be maladaptive or potentially detrimental in the context of immune-mediated and chronic inflammatory diseases [10, 74, 75].

INFECTIONS

Macrophage can be trained to play an effective protective role in reinfection. In a mouse model of recurrent skin and skin structure infections, macrophage memory reduced the severity of skin damage and promoted an increase in the production of cytokines associated with skin protection and prevention of dissemination. And the adoptive transfer of trained macrophages into naïve skin has also been established to afford protective efficacy [76]. Furthermore, trained macrophages of mice that have previously been infected with *S. aureus*, were observed to be associated with more rapid monocyte recruitment, enhanced bactericidal effects, improved healing, and enhanced resistance of these mice to secondary infections [77]. However, re-exposure to pathogens can also lead to an immune deficiency state [78]. Patients with sepsis developed a state of immune tolerance and immunosuppression in the late stage, and they were highly susceptible to pathogenic infection and fatal complications, manifested by significantly decreased levels of pro-inflammatory cytokines IL-1 β , IL-6 and TNF- α , which is attributed to peripheral monocytes memory [79].

Atherosclerosis

In addition to microbial products, endogenous non-microbial stimulus ox-LDL and lipoprotein (a) also induce monocyte/macrophage memory and play important roles at different stages of atherosclerosis [31, 80]. In this context, trained human monocytes have been found to switch to an enduring pro-atherogenic macrophage phenotype via epigenetic histone modifications associated with the downregulation of H3K4me3 on the promoters of pro-inflammatory cytokines and a metabolic shift towards increased glycolysis. When stimulated secondarily by TLR2 and TLR4 agonists, there was a

substantial increase in the production of pro-atherogenic factors, such as TNF- α , IL-1 β , IL-6 and MCP-1, which may lead to arteriosclerotic plaque instability and exacerbate atherosclerosis [31, 81–83]. In addition, macrophages also showed glycolytic metabolic reprogramming that relies on HIF-1 α and 6-phosphofructo-2-kinase [84].

Inflammatory diseases

The distinctive feature of inflammatory diseases is the over-production of IL-1 β . IL-1 β , as a potent pro-inflammatory cytokine, also serves as an inducer of immunological memory. For example, IL-1 β pre-treatment was demonstrated to protect mice against lethal *Pseudomonas* infection [85]. Human monocytes primed by IL-1 β in vitro enhanced the production of the pro-inflammatory cytokines IL-6 and TNF- α upon LPS restimulation, along with the upregulation of H3K4me3 levels. Additionally, pre-treatment with the histone methyltransferase inhibitor abrogated the IL-1 β -induced trained immunity response, thereby indicating that this cytokine may play an important role in the macrophage memory associated with inflammatory diseases [29, 85, 86]. This supposition is indeed supported by the fact that β -glucan-induced macrophage memory can contribute to inhibiting IL-1 β -mediated inflammation. β -glucan is recognized as a classical immune memory inducer that can inhibit NOD-, LRR-, and pyrin domain-containing protein 3 (NLRP3) inflammasome, the biochemical function of which is to activate cysteine-requiring aspartate proteinase-1 (caspase-1) and subsequent IL-1 β production, thereby establishing macrophage memory and marked attenuation of IL-1 β secretion in NLRP3-associated inflammatory diseases [87, 88].

Cancer

Innate immune memory has also been established to play a central role in tumour progression. For example, β -glucan-induced mouse granulocyte immune memory was reportedly associated with diminished tumour growth, with this anti-tumour effect being attributed to a reprogramming of neutrophils towards an anti-tumour phenotype, a process dependent on type I interferon signalling, although independent of host adaptive immunity [89]. Indeed, macrophages may play opposite roles by displaying both tumour-suppressing and tumour-promoting properties. For example, in terms of suppression, BCG treatment has long been known to eradicate malignancies such as bladder cancer and melanoma and reduce the risk of leukaemia and lymphoma [90–93]. Furthermore, β -glucan has also been used in clinical trials for cancer therapy such as that for non-small cell lung,

breast, and colorectal cancers [94]. The findings of further studies have revealed that such anti-tumour effects may be attributable to the non-specific protection triggered by epigenetic reprogramming and are associated with the induction of monocyte/macrophage memory [30, 95].

With respect to the tumour-promoting effects of macrophage memory, tumour-associated macrophage (TAM), which serves as important immune regulatory element in tumour progression, has been shown to be involved in the development of an immunosuppressive tumour microenvironment and facilitate tumour immune escape, growth, and metastasis [96, 97]. The salient feature of TAM differentiation is epigenetic reprogramming involving histone modification, such as H3K4me3 and H3K9me3 alterations in the promoter regions of IL-6 and TNF- α , thereby inducing the production of pro-inflammatory cytokines and expression of tumour-associated gene profiles [98]. Furthermore, the tumour microenvironment induces macrophage reprogramming to counteract the anti-tumour effects and promote the transition of a chronic inflammatory state, which is associated with the promotion and maintenance of disease progression and neoplastic transformation [99].

Other diseases

Human and mouse macrophages trained by low-dose LPS attenuated the inflammatory phenotype of endometriotic cells via an IL-10-dependent pathway, and significantly upregulated H3K4me3 on the IL-10 promoter and altered inflammatory cytokines production have been observed simultaneously in endometriosis [100]. Moreover, evidence of macrophage memory has also been obtained with respect neurodegenerative disorders, such as Alzheimer's disease, Parkinson's disease, and autism spectrum disorder [75]. In a mouse model of Alzheimer's pathology, LPS-induced immune training has been demonstrated to exacerbate cerebral β -amyloidosis, whereas this could be alleviated by immune tolerance, attributable to microglial (brain-resident macrophage) memory, along with altered inflammatory cytokines production and H3K4me1 and H3K27ac enrichment [101]. And the functional changes in microglia were also accompanied by epigenetic reprogramming at the HIF1A gene locus, consistent with peripheral-trained immunity [53]. Furthermore, in respiratory diseases such as allergic asthma, allergen-triggered macrophages have been observed to acquire a TNF-dependent inflammatory memory and show an excessive mediator response upon stimulation, thereby resulting in a shift towards an M2-like macrophage phenotype and the production of diverse inflammatory

cytokines. Conversely, TNF blockade or genetic ablation has been found to prevent the inflammatory imprinting of macrophages [102].

Conclusion and perspectives

Trained or primed macrophages exhibit immune memory similar to that of adaptive immunity when challenged with secondary triggers. There are two types of macrophage memory, namely, trained immunity and endowed immunity. Trained macrophages acquire a memory phenotype via epigenetic and metabolic reprogramming and facilitate an altered immune response upon secondary stimulation. Similarly, primed macrophages can acquire a memory phenotype, which leads to the development of endowed immunity and allogeneic rejection in response to stimulation with the same allogeneic antigen. The discovery of macrophage memory has challenged traditional dogma regarding the limitations of immunological memory and enriched our understanding of macrophage-mediated immune responses. More importantly, unravelling macrophage memory provides a more comprehensive perspective on the pathogenesis of immune-mediated diseases and may lead to establishing a new class of vaccines and the development of novel immunoprophylactic and therapeutic strategies.

Nevertheless, some questions remain outstanding and the critical future lines of research need to explore: How long can macrophage memory last? Because efficient long-term immune memory is necessary to ensure the durable responses of pharmacological treatments directed against diseases. What is the material basis for inducing macrophage memory given that it only responds to partial stimulation? Is it the result of infection, vaccination, or sterile triggers? Furthermore, it is unclear how different stimulations affect macrophage memory. And there is still much work to be done to fully understand the molecular mechanisms of macrophage memory. More importantly, the role of macrophage memory in diseases needs to be further explored. Recent studies have found that β -glucan treatment of monocytes from volunteers with experimental endotoxemia restored their capacity for cytokines production, and about 60% of distal element histone modification levels of tolerant genes have been reversed [103]. Additionally, β -glucan treatment was also reported to reverse the tumour immune microenvironment, and promoted the activation of M1 macrophages and the production of pro-inflammatory cytokines, thereby facilitating the induction of anti-tumour immunity [94]. Therefore, treatment options based on inducing or regulating macrophage memory warrants further studies.

AUTHOR CONTRIBUTIONS

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The authors have declared that no competing interest exists.

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